



Patient Controlled Intravenous Analgesia Procedure

1. Applicability

Site	Operational Area	Applicable to
Armadale Kalamunda Group	All inpatient services	Medical and Nursing

2. Purpose

The purpose of this procedure is to ensure safe and accurate care is provided to patients who are receiving analgesia through a patient-controlled analgesia device (CADD pump) within Armadale Kalamunda Group (AKG).

It is a legal requirement that all Registered Nurses, Midwives, Nurse/Medical Practitioners or Anaesthetic Technicians who prescribe, store, dispense or administer medications via CADD pump comply as stated with the following legislative requirements:

- [Medicines and Poisons Act 2014](#)
- [Medicines and Poisons Regulations 2016](#)
- [Pharmacy Act 2010](#)
- [Health Practitioner Regulation National Law \(WA\) Act 2010](#)
- Health Department of Western Australia (HDWA) [Prescription and Management of Patient Controlled Intravenous Analgesia \(OD 0658/16\)](#)

3. Background

Patient controlled intravenous analgesia (PCIA) refers to a method allowing a person in pain to deliver their own pain relief. Prescription of a small bolus dose, use of “lockout” intervals and strictly patient use only contribute to the safety of the technique. It is used to manage moderate to severe pain, when oral or intermittent analgesia would be less effective.

The PCIA is prescribed by an anaesthetist. The pump is programmed by either the prescriber/Anaesthetic Technician under the direction of the Anaesthetist or a competent Theatre Registered Nurse (RN); two RNs are required to program the PCIA together. The same two persons who prepare the PCIA are then required to attach the PCIA to the patient. As per Armadale Kalamunda Group [Medication Management Procedure](#).

At Armadale Health Service, the Acute Pain Service (APS) manages these patients and has pre-programmed pumps which accept pre-mixed bags of Fentanyl which should be used when available. Other PCIA therapies such as Morphine or HYDROMORPHONE require preparation by two medication competent staff.

4. Prescription requirements

A PCIA order should only be prescribed or altered by staff deemed competent to do so. At AHS persons authorised to prescribe are as follows: An anaesthetist; or an intensive care or emergency medicine practitioner in consultation with an anaesthetist on a hospital approved prescription chart.

Note: PCIA Plus continuous infusion is only for use in IC

4.1 PCIA Prescriptions should include:

- Date and time
- Drug concentration (mg/mL or mcg/mL)
- PCIA bolus dose (mg or microg)
- Maximum hourly dose (mg or mcg) (This is recommended but not mandated)
- Lockout (min)
- A PCIA order should be prescribed as part of a multimodal analgesic regimen including regular paracetamol and/or NSAIDs (non-selective or COX-2 selective) and/or other adjuvants, bearing in mind contraindications.
- All patients with a PCIA should be referred to and managed by the APS.
- There should be at a minimum one daily medical review of patient and prescription.
- Any changes to a PCIA prescription need to be authorised by the APS/Duty Anaesthetist (DA).
- No additional opioids / sedatives should be given by any route unless ordered by the authorised prescriber. 'Not to be given with PCIA' should be written against any other opioid or sedative prescription, as indicated, for clarity.
- Where multiple drugs are prescribed for analgesia, it is recommended that either these are charted on the same order / medication chart or documented in a manner which allows co-administered drugs to be easily reviewed. Where standard medication charts are used in conjunction with the PCIA chart, a notice alerting users to the PCIA prescription should be inserted, e.g., "PCIA chart in use" alert in the standard drug chart, or "other drug" alert on the PCIA chart. This may include notification that the prescribing APS or duty anaesthetist should be consulted if analgesic prescriptions are to be altered.
- Where other drugs with sedative potential are prescribed, these should be readily apparent upon review of the medication chart.

4.2 Contraindications and special precautions

The following points should be considered by the prescriber and RN (administering the treatment) before commencing the PCIA:

- Physical or mental barriers, such as spinal cord injuries, cognitive impairment
- Concurrent sedative use
- Airway compromise
- Acute alcohol intoxication
- Obstructive sleep apnoea / COPD / asthma which may increase the risk of respiratory depression
- Morbid obesity
- Renal impairment
- Previous allergy / sensitivity to a specific opioid

4.3 Competency Requirements

- The PCIA is to be set up, prepared, checked and commenced by two competent AHS staff.

- Roles delineated for all staff, please refer to the [AKG Medication Management Procedure](#).
- The patient with a PCIA is to be cared for by a Registered Nurse / Midwife/ Enrolled Nurse who holds a PCIA competency.
- PCIA competency is obtained by:
 - Successful completion of the IV PCIA E-Learning package and follow up demonstration of competency following PCIA Competency checklist; or
 - Bedside teaching of the PCIA device and competency requirements, with follow up completion of the learning package.
- Registered Nursing or Midwifery staff caring for the patient must work within the Australian Nursing and Midwifery Council (ANMC) scope of practice decision making framework (DMF).

4.4 Patient safety

- The Patient Controlled Intravenous Analgesia & Ketamine Infusion Record (AKMR 177) must be used to record the number of patient attempts and delivered drug doses for all patients with a PCIA.
- Naloxone 400mcg must be available on the resuscitation trolley in an environment where a PCIA is utilised.
- Whilst on a PCIA the patient is not to leave the ward area unless accompanied by a RN.

4.5 Equipment

- The bag must be changed when empty (if it is still clinically required) or every 72 hours when the intravenous lines are changed. The bag must ONLY be changed by two nurses who have gained competency in PCIA (one must be a RN).
- A PCIA giving set with anti-siphon and anti-reflux valves must be used.
- The infusion line should be clearly labelled with IV opioid labels and colour coded according to recognised standards.
- PCIA pumps should be locked to avoid reprogramming or tampering.
- The pump should be checked at the beginning of each shift change and whenever the PCIA pump is opened (two nurses are required if the prescription is being changed). Once completed this should be documented on the PCIA chart.
- ters detailed on the Adult Observation & Response Chart (AKMR140.3).
- Additional patient observations must be performed following initiation of the PCIA and recorded on the PCIA Chart (AKMR 177).

4.6 Oxygen and Pulse Oximetry

- Oxygen (minimum 6L/min via face mask or 2L/min via nasal prongs) may be prescribed for the duration of therapy or as otherwise directed by the prescribing medical practitioner to maintain Oxygen Saturation at either 92-94% or >94%, as per individual patient target saturation percentage.
- In line with the Operational Directive Guidelines for Acute Oxygen Therapy for Western Australian Hospitals, the prescribing medical practitioner must specify the target saturation percentage for each patient.

4.7 Discontinuation of PCIA

- Most patients self-wean off their PCIA as their pain decreases.

- Aim to discontinue PCIA during daytime hours, ideally 0600hrs
- Patient should already have started alternative delivery (e.g. oral) analgesia (e.g. paracetamol / NSAIDs / oral opioid).
- Any remaining opioid must be disposed of according to hospital policy and in line with the [Medicines and Poisons Act 2014](#).
- Refer to [Medication Management Procedure \(AKG-PRO-0255-18\)](#).

5. Compliance monitoring

Compliance against this policy will be evaluated with completion and investigation of CIMS whenever a clinical incident resulting in actual or potential harm to a patient occurs. PCIA prescription charts and Adult Observation & Response Chart (AKMR140.3) will be reviewed during the APS rounds daily to check that documentation is complete and accurate.

6. Policy context

The following documents are required to give effect to this procedure:

- [Medicines and Poisons Act 2014](#).
- [Medicines and Poisons Regulations 2016](#)
- [Pharmacy Act 2010](#)
- [Health Practitioner Regulation National Law \(WA\) Act 2010](#)
- [Medication Policy \(AKG-POL-0253-13\)](#)
- Health Department of Western Australia (HDWA) [Medication Review Policy MP 0104/19](#)
- HDWA [Medication Chart Policy MP 0078/18](#)
- HDWA [National Standard for User Applied Labelling of Injectable Medicines Fluids and Lines OD 0647/16](#)
- HDWA [Statewide Medicines Formulary Policy MP 0077/18](#)
- HDWA [Reporting of Schedule 4 Restricted and Schedule 8 Medicines Discrepancies Policy \(MP 0103/19\)](#)
- HDWA [Prescription and Management of Patient Controlled Intravenous Analgesia \(OD 0658/16\)](#)
- [WA Health Storage and recording of Restricted Schedule 4 \(S4R\) medicines OD 0528/14](#)
- [HDWA High Risk Medication Policy OD 0561/14](#)

7. Supporting information

The following documents support the implementation of this procedure:

- [HDWA Best Practice Principles for Medication Review Guidance Document](#)
- [HDWA Prescription and Management of Patient Controlled Intravenous Analgesia OD 0658/16](#)
- East Metropolitan Health Service (EMHS) [Patient Identification and Procedure Matching \(EMHS: 21\)](#)
- [HDWA Statewide Medicines Formulary Guideline](#)
- [Medication Management Procedure \(AKG-PRO-0255-18\)](#)
- [Hand Hygiene Policy \(AKG-POL-0187-17\)](#)

8. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 3 – Action 3.8
- Standard 4 – Action 4.3; 4.4; 4.10
- Standard 8 – Action 8.4; 8.6

9. References

1. Australian Nursing and Midwifery Council (2007) National Framework for the development of decision making tools for nursing and midwifery practice, * rebranded effective (2013) Canberra
2. Government of Western Australia State Law Publisher. Medicines and Poisons Regulations 2016
3. Nurses Board of Western Australia. (2002) Report of the Scope of Nursing Practice Project. Perth: Nurses board of Western Australia
4. State Law Publisher, Nurses and Midwives Act 2006, Western Australia
5. Australian Nursing and Midwifery Council (2015) Framework for assessing standards for practice for registered nurses, enrolled nurses and midwives. Nursing and Midwifery Board of Australia, Canberra
6. National Safety and Quality Health Service Standards, Guide for Hospitals. (2017), Australian Commission on Safety and Quality in Health Care
7. ANMJ Australian Nursing & Midwifery Journal. (2019) Patient Controlled Analgesia, ANMJ, Australia <http://anmj.org.au/patient-controlled-analgesia/> <http://anmf.cliniciansmatrix.com>

10. Document contact

Enquiries relating to this procedure may be directed to:

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11. Review

This procedure will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V4.1	25/05/2023	25/05/2026	Minor amendment

12. Authorisation

This procedure has been authorised and issued by:

Authorised by	Dr Parr – Director Clinical Services
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